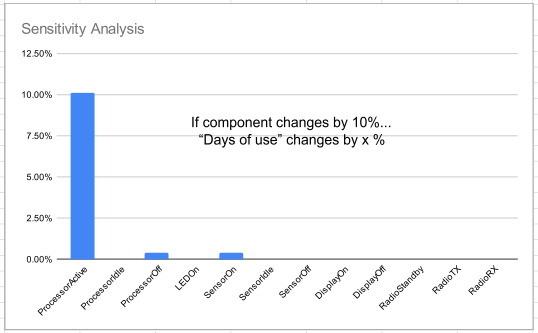
System Parameters (defined by hardware)			Profiles (usage of each component mode - defined by software and usage)			
form the datasheets						
			"off"	"sensing"	"interactive"	
Processor						
Active	530	mW	0%	0%	100%	
Idle	50	mW	0%	100%	0%	
Sleep	0	mW	100%	0%	0%	
LED						
On	0	mW	0%	0%	0%	
Sensor						
On	0	mW	0%	0%	0%	
Idle	0	mW	0%	0%	0%	
Off	0	mW	0%	0%	0%	
Display						
On	1200	mW	0%	0%	100%	
Off (leakage)	0	mW	0%	0%	0%	
Radio						
Data Rate	0	bps	0%	0%	0%	
Standby Power	0	mW	0%	0%	0%	
TX Power	0	mW	0%	0%	0%	
RX Power	0	mW	0%	0%	0%	
			23.95	0	0.05	hours/day typical usage
Battery						
Capacity	2000	mAh				
Nominal Voltage	4	V				
Regulator Efficiency	90%					



Total power in profile (mw)			Maximum Time	
"off"	0.1	mW	66600.0	hours
"sensing"	50	mW	133.2	hours
"interactive"	1730	mW	3.8	hours
Effective Battery Capacity				
	6660	mW*h		
Days of Use	**74.92**	**days**		
Hours of Use	**1798.08**	**hours**		

REFLECTIONS : WHAT DID YOU LEARN FROM ANALYZING YOUR POWER. TALK ABOUT SOME POTENTIAL TRADEOFFS.

Reflections on Power Analysis:

1. Determining the "Days of Use" Metric:

I calculated the usage based on a scenario of 20 visitors per day. For the Sensing Device, I assumed each event takes 10 seconds to wake up, process the vibration, and transmit data via WiFi. This results in a very low duty cycle where the device spends 99.9% of its time in Deep Sleep.

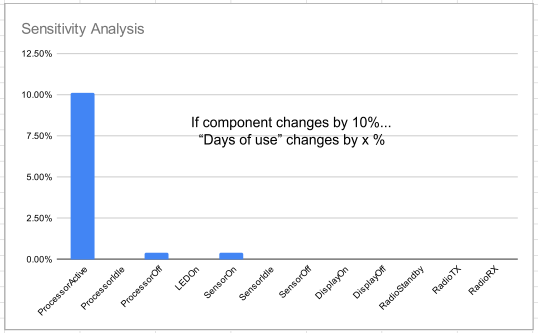
2. Optimum Battery Size:

For the Sensing Device, a 1000mAh LiPo is the optimum size because it is thin enough to fit inside a small door-mounted enclosure while providing over 50 days of battery life. For the Display Device, a 2000mAh battery or direct USB power is better since the stepper motor consumes significant power (1200mW) during interaction.

3. Hardware/Software Tradeoffs:

To improve the user experience and battery life, I could implement the ESP-NOW protocol instead of standard WiFi to reduce the handshake time from seconds to milliseconds. Another tradeoff would be using a low-power E-ink display instead of a motor, but that would lose the "calm tech" physical interaction which is a key design goal of this project.

System Parameters (defined by hardware)			Profiles (usage of each component mode - defined by software and usage)			
form the datasheets						
			"off"	"sensing"	"interactive"	
Processor						
Active	530	mW	0%	0%	100%	
Idle	50	mW	0%	100%	0%	
Sleep	0	mW	100%	0%	0%	
LED						
On	0	mW	0%	0%	0%	
Sensor						
On	0	mW	100%	100%	100%	
Idle	0	mW	0%	0%	0%	
Off	0	mW	0%	0%	0%	
Display						
On	0	mW	0%	0%	0%	
Off (leakage)	0	mW	0%	0%	0%	
Radio						
Data Rate	0	bps	0%	0%	0%	
Standby Power	0	mW	0%	0%	0%	
TX Power	0	mW	0%	0%	0%	
RX Power	0	mW	0%	0%	0%	
			23.9	0	0.1	hours/day typical usage
Battery						
Capacity	1000	mAh				
Nominal Voltage	4	V				
Regulator Efficiency	90%					



Total power in profile (mw)			Maximum Time	
"off"	0.2	mW	16650.0	hours
"sensing"	50.1	mW	66.5	hours
"interactive"	530.1	mW	6.3	hours
Effective Battery Capacity				
	3330	mW*h		
Days of Use	**57.62**	**days**		
Hours of Use	**1382.94**	**hours**		

REFLECTIONS : WHAT DID YOU LEARN FROM ANALYZING YOUR POWER. TALK ABOUT SOME POTENTIAL TRADEOFFS.

Reflections on Power Analysis:

1. Determining the "Days of Use" Metric:

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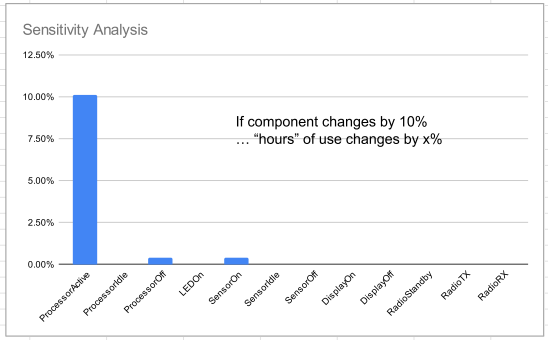

Linked Data from First Page DO NOT EDIT

System Parameti															Profiles (usage o				
	form the datashe																		
															"off"	"sensing"	"interactive"		
Processor																			
Active	530	mW	477	530	530	530	530	530	530	530	530	530	530	530	0%	0%	100%		
Idle	50	mW	50	45	50	50	50	50	50	50	50	50	50	50	0%	100%	0%		
Sleep	0	mW	0	0	0.09	0	0	0	0	0	0	0	0	0	100%	0%	0%		
LED																			
On	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
Sensor																			
On	0	mW	0	0	0	0	0.09	0	0	0	0	0	0	0	100%	100%	100%		
Idle	0	mW	0	0	0	0	0	0.09	0	0	0	0	0	0	0%	0%	0%		
Off	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
Display																			
On	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
Off (leakage)	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
Radio																			
Standby Power	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
TX Power	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
RX Power	0	mW	0	0	0	0	0	0	0	0	0	0	0	0	0%	0%	0%		
															23.9	0	0.1	hours/day typical usage	
Battery																			
Capacity	1000	mAh																	
Nominal Voltage	4	V																	
Regulator Efficie	90%																		
																#REF!	#REF!		
																#REF!	#REF!		

																				#REF!	#REF!		
REFLECTIONS :		#REF!	#REF!											#REF!	#REF!	#REF!				#REF!	#REF!		
Total power in pr																							
"off"	0.2	mW	0.2	0.2	0.19	0.2	0.19	0.2	0.2	0.2	0.2	0.2	0.2	0.2									
"sensing"	50.1	mW	50.1	45.1	50.1	50.1	50.09	50.1	50.1	50.1	50.1	50.1	50.1	50.1									
"interactive"	530.1	mW	477.1	530.1	530.1	530.1	530.09	530.1	530.1	530.1	530.1	530.1	530.1	530.1									
Effective Battery																							
	3330	mW*h	3330	3330	3330	3330	3330	3330	3330	3330	3330	3330	3330	3330									
Days of Use	57.62242603	days	63.44065536	57.62242603	57.86172265	57.62242603	57.86272806	57.62242603	57.62242603	57.62242603	57.62242603	57.62242603	57.62242603	57.62242603									
Hours of Use	1382.938225	hours	1522.575729	1382.938225	1388.681344	1382.938225	1388.705474	1382.938225	1382.938225	1382.938225	1382.938225	1382.938225	1382.938225	1382.938225									
% change			10.10%	0.00%	0.42%	0.00%	0.42%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%									
Parameter Name			ProcessorActive	ProcessorIdle	ProcessorOff	LEDOn	SensorOn	SensorIdle	SensorOff	DisplayOn	DisplayOff	RadioStandby	RadioTX	RadioRX									